

1RV24MC042_HARSHAVARDHAN_BR_LAB6

Deploy an Automated Build Pipeline Using Docker Hub

Prerequisites:

- Dockerhub, github, docker hub PAT, docker installed locally, git installed locally

Project structure

```
Program-6/  
|-- Dockerfile  
|-- go.mod  
|-- go.sum  
|-- main.go
```

```
# Stage 1: Builder  
FROM golang:1.23 AS builder  
  
WORKDIR /app  
  
# Use direct proxy for reliability  
ENV GOPROXY=https://proxy.golang.org,direct  
ENV GOSUMDB=off  
  
COPY go.mod go.sum ./  
RUN go mod download  
  
COPY . .  
RUN CGO_ENABLED=0 GOOS=linux go build -o app .  
  
# Stage 2: Final image (scratch)  
FROM scratch  
WORKDIR /app  
  
COPY --from=builder /app/app .  
  
EXPOSE 3000  
ENTRYPOINT ["/app"]
```

```
//index.js  
package main  
import (  
    "github.com/gin-gonic/gin"  
)  
  
func main() {  
    r := gin.Default()  
  
    r.GET("/", func(c *gin.Context) {  
        c.String(200, "Hello from Program-6 Docker Automation with Go + Gin!")  
    })  
  
    r.Run(":3000")  
}  
  
});
```

```
module program6  
go 1.22  
require github.com/gin-gonic/gin v1.10.0
```

Step 1 : Create Github PAT

Github -> settings -> developer settings -> personal access tokens

choose classic tokens

generate with : write: package and read:package permissions

save token safely

Step 2 : Create docker file and build Image

```
docker build -t program6 .
```

Step 3 : Tag image for Docker Hub

```
docker tag program6 -u <dockerhub_username>
```

Step 4 : Push image to docker hub

```
docker login -u <dockerhub-username>
```

Step 5: Tag image for github container registry (GHCR)

```
docker tag program6 ghcr.io/<github-username>/program6:latest
```

Step 6 : Push Image to GHCR

Login using github pat

```
echo $GITHUB_PAT | docker login ghcr.io -u <githubusername> --password-stdin
```

```
docker push ghcr.io/<githubusername>/program-6:latest
```

Step 7 : Connect Project to Github

```
git init
git remote add origin https://github.com/<github-username>/program-6.git
git remote -v
```

Step 8 : Push source code to github

```
git add .
git commit -m "Initial commit"
git push -u origin main
```

Step 9 : Verify Uploads

Github

- verify github repo
- check packages : image should appear as ghcr.io/<github_username>/program6 latest
Docker Hub
- Go to repositories
- check and verify



program6 latest

Private

Latest

 Install from the command line

[Learn more about packages](#)

```
$ docker pull ghcr.io/hvardhan1024/program6:latest
```

Last published

11 days ago

Total downloads

0

 **Package settings**

Recent tagged image versions

latest

Published 11 days ago · Digest 

 0

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harshavardhan873/prog6 

Last pushed less than a minute ago ·  0 ·  0

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Tags

 DOCKER SCOUT INACTIVE
[Activate](#)

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
 latest		Image	less than 1 day	less than a minute

[See all](#)

Repository overview INCOMPLETE

An overview describes what your image does and how to run it. It displays in [the public view of your repository](#) once you have pushed some content.

[Add overview](#)

Using 0 of 1 private repositories.

Docker commands

[Public view](#)

To push a new tag to this repository:

```
docker push harshavardhan873/prog6:tagname
```



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